

Jacob Cockerill

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Education:

Master of Science, Mechanical Engineering

Expected May 2028

Texas A&M University, College Station, TX

Bachelor of Science, Mechanical Engineering

May 2026

Rose-Hulman Institute of Technology, Terre Haute, IN

Concentration in Aerospace Engineering

Certification in the Fundamentals of Engineering: Mechanical

Skills:

Software: Windows OS, Microsoft Office Suite, SolidWorks, ANSYS Mechanical (Static Structural, Modal & Harmonic Response), ANSYS Fluent, MEscape, EES, PicoSoft

Programming: MATLAB, Python, C++, Maple

Industry Experience:

Renascent, Inc. (Indianapolis, IN) *Research & Development Engineering Contractor*

June 2026 – August 2026

- Extended the prototype's functionality from the Senior Design Project of requiring a prescribed lift process to supporting variable boom and stick trajectories, including lifts with pauses and restarts, by incorporating acceleration-based motion compensation into the payload estimation model.
- Standardized the prototype's sensor mounts, wiring, data acquisition, computer setup, and software into a plug-and-play installation kit deployed on a CAT 335F and CAT 374F excavators.
- Redesigned the touchscreen's user interface to automate payload tracking, requiring operator input to start a new truck or apply tare corrections, while adding machine and bucket profiles, load history, protected settings, and USB data export.
- Reformulated the prototype's payload estimation process as an inverse surrogate-model problem using hydraulic pressure response and boom and stick positions, with a load-dependent Kronecker-product motion feature derived from boom and stick accelerations.
- Validated the system using bucket-load estimates across 205 lifts on a CAT 335F achieving a 0.065-ton mean absolute error and six truck-load estimates with a 0.35-ton mean absolute error.

Richmond Sanitary District (Richmond, IN) *Engineer Intern*

June 2024 – August 2024

- Developed a spreadsheet system to analyze flow, power, and runtime data across 10 municipal lift stations to establish performance baselines and identify operational inefficiencies using Flowlink Cipher data.
- Verified and corrected as-built drawings against field conditions, updating the district database with unmapped structures.

Winchester Wastewater Treatment (Winchester, IN) *Engineer Intern*

June 2023 – August 2023

- Performed routine inspections of treatment-plant and remote-station pumps and logged operating hours and sensor status for maintenance records.

Selected Technical Projects:

Senior Design Project: Excavator Payload Estimation System for Renascent, Inc.

September 2025 – May 2026

- Designed and built an excavator payload estimation prototype using hydraulic pressure transducers, dynamic inclinometers, a Raspberry Pi, and an in-cab touchscreen for approximately \$2,500 per excavator compared with \$15,000 for the closest similar aftermarket solution.
- Developed a Python calibration process and estimation model for a controlled lift using hydraulic pressure, boom and stick positions, and engine speed to estimate bucket payload without direct geometric measurements.
- Field-validated the prototype across five payloads and six engine speeds achieving a 2.59% mean absolute relative error and 64.9 lb mean absolute error.

Selected Technical Projects Continued:

2D Finite-Volume Solver for Compressible Nozzle Flow

April 2026 – May 2026

- Developed a custom 2D finite volume solver in MATLAB to model steady inviscid compressible flow throughout a converging nozzle.
- Implemented a multi-dimensional finite volume scheme, structured skewed-mesh generation, boundary conditions, and a mesh-refinement analysis to resolve the flow-field solution variables.
- Validated the solver against a quasi-1D isentropic nozzle model, achieving maximum relative errors below 1% in Mach number, pressure, density, and temperature while capturing cross-stream flow variation.

Dynamic Modeling & Simulation of a Cincom D25 VIII CNC Automatic Lathe

January 2026 – February 2026

- Modeled and simulated the secondary tool-post drive system of a Cincom D25 VIII CNC automatic lathe, including servo motors, belt and chain drives, ball screws, tool positioning, and machining loads.
- Created mathematical models using bond graphs for the X2 and Y2 rotary-tool devices and the dual-motor linear positioning system.
- Derived and parameterized the Y2 drive equations for the electrical, rotational, and linear components, including a compliant tool-to-workpiece interface.
- Simulated step, pulse, and ramp inputs up to 24 V in MATLAB and evaluated motor speed, output force, ball-screw position, and motor current.

Fatigue Analysis of a Multi-Step Driveshaft

September 2025 – October 2025

- Modeled combined bending and torsional stresses along a stepped driveshaft to identify critical fatigue locations under steady-state operations.
- Developed fatigue life predictions using both Morrow and Smith–Watson–Topper corrections, selecting models based on material behavior and mean stresses.
- Formulated a nonlinear constrained optimization problem to minimize shaft mass by iterating on section diameters and fillet radii while enforcing fatigue life requirements.
- Applied design constraints such as minimum step changes, notch sensitivity factors, and conservative load factors to ensure a feasible solution.

Topology Optimization of a Complex Truss System

March 2025 – May 2025

- Developed a custom 2D finite element solver in MATLAB to model the stress and strain in a nonlinear truss system under static loading.
- Implemented multidimensional Newton–Raphson to numerically solve the nonlinear system.
- Coupled the numerical solver with a genetic algorithm to perform topology optimization, minimizing material usage subject to displacement constraints and continuity conditions.
- Verified solver accuracy by benchmarking reaction forces and deflections against analytical solutions and linear finite element models across increasing complexity.

Model Predictive Control of a Mass-Spring-Damper System

January 2025 – February 2025

- Designed and implemented a model predictive controller for a nonlinear mass-spring-damper system, incorporating pseudo-linearization for improved accuracy in MATLAB.
- Built a state-space simulation with input and state constraints, solving each receding-horizon step via quadratic programming to enable real-time control.
- Validated the controller through hardware implementation, achieving robust control trajectory tracking despite disturbances.